Exercise 1 SQL fundamentals

01

```
SELECT *
FROM employees;
```

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	~~60 000~~ 45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

02

```
SELECT DISTINCT department
FROM employees;
```

department
IT
HR
Finance
Marketing

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03

```
SELECT  first_name,
        last_name
FROM    employees
ORDER BY salary DESC;
```

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

04

```
SELECT  id,
        first_name,
        last_name,
        salary
FROM    employees
ORDER BY salary DESC
LIMIT 3;
```

id	first_name	last_name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

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05

```
SELECT id,
       first_name,
       last_name,
       department
FROM employees
WHERE department = 'IT';
```

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

06

```
SELECT id,
       first_name,
       last_name,
       department,
       salary
FROM employees
WHERE department = 'Finance' AND salary > 60000
```

id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65 000

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07

```
SELECT  id,
        first_name,
        last_name,
        department
FROM  employees
WHERE  department = 'HR' OR department = 'Marketing';
```

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

08

```
SELECT  id,
        first_name,
        last_name,
        department
FROM  employees
WHERE  department <> 'IT';
```

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

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09

```
SELECT  id,
        first_name,
        last_name,
        department
FROM    employees
WHERE   department IN ('IT','HR','Finance');
```

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10

```
SELECT  id,
        first_name,
        last_name,
        department,
        salary,
        city
FROM    employees
WHERE   department = 'IT' AND salary > 65000 AND city = 'Johannesburg';
```

id	first_name	last_name	department	salary	city
5	Eva	Naidoo	IT	72000	Johannesburg

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